Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology Using Facial Expression Recognition and an NLP-Based Chatbot

COMMON INTEGRATED SOLUTION REPORT

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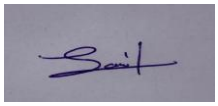
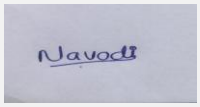
Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. We have disclosed all use of generative AI and other automated tools in accordance with the module policy. We remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this report.

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ABSTRACT

A solution to the potential challenge facing Deaf and hard-of-hearing students presented by online lessons relying on verbal explanations and offering limited opportunities for revisiting information is proposed by SignLearn AI. The integrated learning platform was developed for Sri Lankan Grade 10 and 11 students' ICT course and consists of four research components: attention tracking using knowledge graph-based questioning, an emotion-aware adaptive chatbot, a sign-avatar-based lecture generator, and a smart haptic-wristband with interactive sign practice. The components were unified by the idea of using the context of learning materials for lectures, questions for concept checking, revision, and sign practice.

The components were implemented using a combination of modular and integrated approaches. Technologies included the React front-end, the FastAPI service and MongoDB database, webcam-based attention tracking, knowledge graph mapping for identifying relevant concepts for questions and reviews, contextual chatbot assistance with revision suggestions and local database-driven fallback responses in case of language-model service failures, interactive gesture-based instruction with repayable signs, and camera-based practice and configurable haptic notifications as part of the extended interactive loop.

As a result of the research, a platform was developed that unified the described components and organized the activities around student, lesson, and topic contexts. The report assesses the system using such criteria as interface coherence, reliability of the components, accessibility, and general usability. The provided documentation allows for concluding on the implemented scope but does not provide sufficient evidence of improved efficiency and stakeholder acceptance. Therefore, the SignLearn AI solution should be considered proof-of-concept for building an inclusive system around accessible ICT learning. Its practical application and scientific validity depend on further research, including local sign-language validation and interaction studies with the target audience.

Keywords: Accessible learning, Deaf and hard-of-hearing learners, Adaptive learning, Knowledge graphs, Sign-language support

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASL	American Sign Language
BLE	Bluetooth Low Energy
BSL	British Sign Language
EAR	Eye Aspect Ratio
EARA	Emotion-Aware Response Adaptation
FER	Facial Expression Recognition
G.C.E. O/L	General Certificate of Education Ordinary Level
GQSA	Graph-Based Question Selection Algorithm
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICT	Information and Communication Technology
IoT	Internet of Things
JSON	JavaScript Object Notation
LLM	Large Language Model
LSTM	Long Short-Term Memory
MAR	Mouth Aspect Ratio
MCQ	Multiple Choice Question
MLP	Multi-Layer Perceptron
MNIST	Modified National Institute of Standards and Technology
NLP	Natural Language Processing

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Digital learning provides students with access to lessons, practice questions and self-learning. However, deaf and hard-of-hearing students may encounter barriers when the learning material uses audio explanation and notifications. The World Health Organization has identified access to written information, captions and national sign-language interpretation as measures beneficial to communication and learning [1]. The guidance from the World Wide Web Consortium (W3C) also emphasizes the importance of captions, and the fact that learners have diverse degrees of use of written and signed language [2], [3].

The research focuses on deaf and hard-of-hearing students studying the Sri Lankan Grade 10 and 11 Information and Communication Technology (ICT) syllabus for the G.C.E. O/L examination. These students are the primary stakeholders, supported by ICT teachers, special education instructors, schools and parents or guardians. The intended setting involves classroom learning and self-study, where students engage with video lessons, explanations and practice activities to develop their understanding of ICT concepts.

The common problem involves both access to lesson content and support when learning difficulties arise. A student who has missed an explanation may require help identifying the relevant lesson segment, revisiting prerequisite concepts and practicing the topic. Teachers also need understandable information about these difficulties to provide further support. Accessible content and relevant revision opportunities are therefore central to the project.

SignLearn AI addresses this problem with four connected components: attention monitoring with knowledge graph-based questions, an emotion-aware adaptive chatbot, a sign-avatar lecture generator and a smart haptic wristband with sign-language practice. Across the platform, lessons, recommendations and sign activities are organized around Grade 10 and 11 ICT topics. The research looks into how these components can work together to support accessible learning, while using attention observations alongside learner feedback and question responses rather than treating them as definite evidence of understanding.

1.2 Stakeholders and user needs

The primary stakeholders for SignLearn AI are the deaf and hard-of-hearing students, who are studying Grade 10 and 11's ICT. These students require access to lesson content that can be understood and explained in detail, as well as provide the opportunity for revisions of the challenging topics at their own pace. As there may be a range of communication preferences, the lessons should include transcripts, explanations, demonstrations, and notifications in an appropriate format beyond auditory information.

ICT teachers and special education instructors are the secondary stakeholders for the project. They are likely to be concerned with the progress made by the students and the ability to obtain the necessary information for additional assistance. Schools will be focused on the creation of a useful tool that is not required to seek external support for its proper functioning, while parents and guardians can be expected to aid in the child's independent learning. These stakeholders' concerns are summarized in Table 1.1 below.

Table 1.1: Stakeholders, User Needs and Platform Support

Stakeholder	**Main user needs**	**Platform support**
Deaf and hard-of-hearing students - Primary	Accessible explanations, topic-specific revision, sign practice and non-audio alerts	Transcripts, adaptive chatbot, syllabus-based questions, sign demonstrations and wristband feedback
ICT teachers and special education instructors - Secondary	Identify difficult concepts and guide individual learners	Learning analytics, answer histories, repeated-query alerts and attention summaries
Schools and educational institutions - Secondary	Curriculum alignment and practical use with available equipment	Grade 10 and 11 topic organisation and configurable learning services
Parents and guardians – Secondary	Support home learning and understand areas requiring assistance	Progress information that can be discussed with students and teachers
System administrators - Supporting	Maintain accounts, learning resources and system operation	Account management, content management and service configuration

The needs identified in Table 1.1 can be seen as stemming from the project scope and accessibility considerations discussed in 1.1. They are presented as design requirements rather than as findings from actual stakeholder engagement.

An example of a user using such a system might be a Grade 10 pupil having watched a lesson on computer hardware and requiring additional explanation. They could look at the transcript and ask for a less complex chatbot explanation and relevant sign demonstration. A teacher could use such a feature to assess student attempts at explaining and learning by reviewing their attempts and asking whether further assistance is required.

1.3 Alignment with the Sustainable Development Goals

SignLearn AI mainly focuses on Sustainable Development Goal (SDG) 4, which is about ensuring a quality education for everyone. Specifically, Target 4.5 requires that education be made accessible to all vulnerable groups, including those with disabilities, while Target 4.a demands that education systems be made more inclusive through addressing the needs of persons living with disabilities (4).

The project contributes to these targets by providing teaching assistance to deaf and hard-of-hearing students who are studying in Grade 10 and 11 in the topic of ICT. Supplementary transcripts, contextual explanations and demonstrations, and questions based on a knowledge graph all contribute to students' ability to access the same information as their hearing peers. Furthermore, the revision and personalization features allow the users to re-visit the concepts that they have learned and demonstrate their understanding, while the smart bracelet provides tactile feedback that does not require auditory processing. Table 1.2 below contains the analysis of the intended contribution of the project to the targets.

Table 1.2: Alignment of SignLearn AI with SDG 4

SDG target	Intended project contribution
4.5 - Equal access to education	Supply curriculum-based explanations, revision and sign support that tackles accessibility barriers experienced by deaf and hard of hearing learners.
4.a - Inclusive learning environments	Offer textual, visual and tactile support as well as progress information that helps teachers to guide learners.

These contributions are what the project is all about. Their manifestation will depend on the accessibility, accuracy, and usefulness of the combined system. Enhancements in the learning or participation levels have to be evaluated with the students and teachers rather than simply inferred from the given attributes.

1.4 Purpose and scope of the common report

This common report aims to present the common research problem and integrated solution developed through the SignLearn AI. The report discusses how the platform combines four research components to address the needs of deaf and hard-of-hearing students studying the Sri Lankan Grade 10 and 11 ICT syllabus: attention monitoring via knowledge graph-based questions, emotion-aware adaptive chatbot, sign avatar lecture generator, and a smart haptic wristband with interactive sign-language practice.

The report focuses on the general objectives, the system's architecture and the component's interaction, shared data and end-to-end learning workflows. In addition, the report discusses the development and integration process, system-level testing, evaluation, limitations and potential for sustainable adoption. Particular attention is

paid to how data from one component are used in another, such as using attention records to recommend relevant revision or connecting sign practice with haptic feedback.

The four individual theses provide the research and development evidence for each member's component, including the relevant literature, methods, implementation decisions and component-specific evaluation. This common report brings those contributions together to examine how the complete platform addresses the common problem and which outcomes are supported by the available evidence.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

Deaf and hard-of-hearing students learning the Sri Lankan Grade 10 and 11 ICT syllabus require accessible lesson content and appropriate support when encountering learning difficulties. Spoken explanations and audio notifications may restrict access when suitable alternatives are unavailable. WHO and W3C guidance highlight the importance of written information, captions and appropriate sign-language support [1]–[3].

Access to content, however, is only part of the learning challenge. When a student misses a lesson segment or struggles with a concept, they may need help identifying what to revisit and how it relates to earlier knowledge and understanding. For example, a student may struggle to answer a question about logic gates because they missed the current explanation or misunderstood the underlying truth-table notation. Without relevant information about their learning needs, the student may struggle further with an unresolved learning gap, while the teacher has limited information about where assistance is required.

The overall problem addressed by this research is the difficulty of providing deaf and hard-of-hearing Grade 10 and 11 ICT learners with accessible, curriculum-relevant support for recognizing and addressing missed or poorly understood lesson concepts during digital learning.

This problem concerns both the learner's ability to access explanations and the continuity of support between lessons, practice and revision. It also considers teachers' need for meaningful information about learning difficulties. Attention changes alone cannot establish a lack of understanding; support needs must be considered alongside question responses, learner feedback and the lesson context.

2.2 Root causes and overall research gap

The problem is compounded by accessibility, the lack of information regarding learning difficulties and a lack of continuity between lessons and revision. Guidance from WHO and W3C highlights the need for accessible communication through text, captions and appropriate sign-language [1]-[3]. However, providing an accessible version of the lesson content does not automatically identify the concepts that a student has missed and the additional explanation that they need.

Part of the issue is the separation between lesson participation and evidence of understanding. Video progress shows how much content has been played, but not the degree of understanding. Observations of attention could indicate intervals to review, and responses to questions add further evidence of difficulties. Both become more useful when linked to the relevant syllabus concept and its prerequisites.

Another issue comes from the relevance of the support offered without considering the current topic or difficulty of the learner. Reciting an explanation or providing a generic revision activity could leave the initial misunderstanding unresolved. Support needs to take into account of the Grade 10 or 11 curriculum context, the student's question and their explicit feedback.

Visual communication and feedback pose further difficulties. Written explanations may need a complementary sign demonstration for some learners, while those demonstrations need clear hand and finger movements and adequate vocabulary. Students practicing signs need understandable feedback on their attempts. Sound based notifications are not capable of the sole means of drawing attention to these learning events.

Table 2.1 shows how these issues inform the four research components.

Table 2.1: Root Causes and Corresponding Research Components

Root cause or limitation	Resulting support need	Research component
Lesson progress and attention observations do not directly establish concept understanding	Connect potentially missed intervals to curriculum concepts and question responses	Attention monitoring and knowledge graph-based adaptive questions
Explanations and revision may lack the learner's topic and difficulty context	Provide curriculum-relevant explanations and personalized revision	Emotion-aware adaptive chatbot
Text and captions may need complementary sign support	Present clear, repayable demonstrations of selected ICT signs	Sign-avatar lecture generator
Audio alerts are inaccessible to some learners, and sign practice requires feedback	Provide tactile notifications and guided practice with attempt feedback	Smart haptic wristband and interactive sign-language practice

The overall gap in research, which this project aims to fill, is understanding how to integrate lesson-level observations, concept-based assessments, adaptive explanations, sign demonstrations and practice feedback into 1 curriculum-aligned learning workflow for deaf and hard-of-hearing O/L ICT students. It's an integration/evaluation gap, and this doesn't mean those functions aren't implemented elsewhere in some form.

The four components tackle different aspects of that workflow. Attention tracking and questions detect possible review needs, the chatbot offers explanation support, the avatar provides sign demonstrations, and the practice + wristband component allows for participation and feedback. The research must demonstrate that these links

offer cohesive and valuable learning assistance, beyond just providing a few features on one platform.

2.3 Subproblem and component mapping

The overall task is multifaceted and can be further decomposed into four sub-tasks, each of which addresses a different aspect of the learning process. These are: identifying potential learning gaps; providing an explanation of possible learning gaps; presenting sign-based lesson content; and facilitating practice and non-audio feedback. Table 2.2 below correlates subtask with research component and owner.

Table 2.2: Subproblem and Research Component Mapping

Subproblem	Research component and owner	Responsibility and boundary
1. Highlighting concepts that may require review. The playback progress alone does not indicate if a student followed along an explanation or has understood the prerequisites.	Attention Monitoring and Knowledge Graph-Based Adaptive Questions - Tharaka W A K N, IT22211200	Records attention indicators, identifies potential missed intervals and selects concept-related questions. It provides evidence for the review decisions but not treating attention as proof of understanding.
2. Provide explanations and revisions to the level of difficulty of the learner. The student may need simpler diction, a prerequisite reminder, or examination-focused guidance of the current concept.	Emotion-Aware Adaptive Chatbot - Kothalawala L S S, IT22185884	Adapts explanations and revision activities based on the curriculum context, learner feedback and available learning records. It uses attention information provided by the first component instead of performing a webcam analysis itself.
3. Provide visual sign-based support on ICT content. Written explanations may require a supplementary demonstration of relevant signs that the students can replay.	Sign Avatar Lecture Generator - Mendis L B L, IT22197764	Converts the supported lesson terms into sign-avatar sequences and provides lecture playback. Its primary responsibility is to present demonstrations instead of a student practice attempt evaluation.
4. Supporting sign practice and accessible learning alerts. Students require feedback when practicing signs, and a method to receive selected alerts	Smart Haptic Wristband and Interactive Sign-Language Practice - Nawarathna I K K R	Evaluates the camera-based practice attempts, records course progress and delivers configurable tactile alerts. It utilizes sign demonstrations and learning events provided by the

without the use of sound.	B, IT19131948	other components.

The components are interdependent since merely defining the possible challenge is insufficient for providing further assistance. The attention records and concept-question responses can inform the chatbot revision recommendations, while the sign demonstrations give the additional meaning to selected ICT terms, and practice interface allow learners to perform signs and obtain feedback. The wristband notifies the user of the events via tactile alerts.

Within the scope of the project, all components are required to address the described flow of actions since cutting on any of them will result in the loss of essential functions. Specifically, without the first component, the support would be less relevant since chatbots would not collect the necessary data to assess missed concepts. With the second component, detecting the challenges would be insufficient for adjusting the communication method. The third component is indispensable for providing the sign demonstration function, while the fourth component would eliminate the guided practice and tactile notifications. Finally, using the same Grade 10 and 11 topics for the chatbot and wristband unite their scope of research, while the segregation of responsibilities remains apparent.

2.4 Overall aim and group objectives

The overarching purpose of this research is to develop and evaluate a learning platform that provides accessible learning resources and enables deaf and hard-of-hearing students to identify concepts that need to be reviewed during their study of Grade 10 and 11 ICT.

The group objectives in Table 2.3 reflect this purpose as outcomes that can be measured. The component numbers relate to the four research components defined in Section 2.3.

Table 2.3: Group Objectives and Evaluation Evidence

ID	Measurable group objective	Relevant components	Measures and evaluation evidence
1	Organise lesson content, questions, chatbot assistance, and sign activities around common Grade 10 and 11 ICT categories.	All four components	Curriculum-mapping coverage and correctness, determined by content-mapping matrix, example learning flows and teacher examination.
2	Associate potentially missed lesson timing intervals with relevant ICT concepts and provide concept-related questions or recommendations for revision.	1 and 2	Timestamp-into-concept mapping precision, question relevance and successful recommendation retrieval, determined by annotated lesson scenarios and integrated logs.

3	Supply explanations and revision activities that are dependent on adequately addressing the selected topic, learner requests, and feedback.	2, supported by 1	Response correctness, curriculum relevance, and adaptation appropriateness, determined by a structured question set, reviewer scores and stored response comparisons.
4	Pair sign demonstrations with practice activities so that learners could view supported ICT signs, attempt them and receive feedback.	3 and 4	Demonstration clarity, practice-feedback agreement with expert adjudication and progress-record accuracy, determined by examined sign samples and end-to-end practice assessments.
5	Deliver configured haptic notifications for selected learning events and maintain consistency in records across different interactions.	4, integrated with 1–3	Alert-delivery success rate, delivery delay, correct pattern selection and record consistency, determined by device observations and timestamped event logs.
6	Test the usability, accessibility, and reliability of the entire learning experience and consider relevant failure scenarios.	All four components	Completion rate, usability ratings, accessibility observations, integration-test pass rate and recovery performance, determined by user sessions and system-test records.

These objectives relate both to individual contributions and their relationships with one another. For example, a successful revision workflow involves a correct identification of the lesson concept, which is then transferred to the chatbot and used to provide relevant support. Similarly, a sign-practice workflow requires an appropriate demonstration, meaningful feedback on the attempt and accurate recording of progress.

The evaluation framework in Chapter 6 will outline the assessment conditions, sample sizes, baselines and acceptance thresholds for these measures. The achievement of each objective will be inferred from the evidence provided; the mere presence of a feature would not be taken to indicate its effectiveness.

2.5 Scope assumptions and constraints

SignLearn AI focuses on supporting deaf and hard-of-hearing students in studying the Sri Lankan Grade 10 and 11 information communication technology (ICT) syllabus. Its scope includes video lessons and transcripts, attention monitoring, concept-based questions, adaptive chatbot support, sign-avatar demonstrations, sign

practice and haptic notifications. These functions operate within a shared learning platform, with learning records being used to support revision and teacher guidance.

The project covers selected curriculum content and supported ICT sign vocabulary. It does not assume a complete coverage of all syllabus topics, or that all lesson text content can be freely translated into sign language. The platform supports classroom and independent learning, but it does not substitute teachers, qualified sign-language interpreters, or formal examination assessment.

Table 2.4 summarizes the main assumptions, constraints and dependencies of the integrated solution.

Table 2.4: Scope Assumptions, Constraints and Dependencies

Area	**Assumption, constraint or dependency**
Curriculum and learning content	Lessons, transcripts, questions, and revision materials must be correctly categorized under Grade 10 and 11 ICT topics, on which recommendations are dependent.
User equipment	Webcam-based functions require a compatible camera, browser permissions, and enough processing power from the device. Physical haptic feedback requires a connected and powered wristband.
Attention observations	Lighting, camera position, occlusion, and learner movement can affect detection. Attention indicators identify potential review needs and are not direct measurements of emotions or understanding.
Sign-language coverage	Avatar output is limited by the available gesture vocabulary and landmark sequences. WLASL-based signs use American Sign Language data; their appropriateness for the target Sri Lankan learners must be linguistically reviewed.
Sign-practice feedback	Camera-based evaluation depends on visible hands, suitable positioning, and supported signs. Practice scores are system-generated feedback and not certification of sign-language competence.
Chatbot services	Online responses are dependent on the configured language-model service, connectivity, and service limits. The fallback local dataset provides limited support, which is reliant on the available learning materials.

Component integration	Reliable operation depends on consistent student, session, lesson, and concept identifiers, compatible interfaces, and available backend and database services.
Wearable operation	Alert delivery is dependent on device connectivity, firmware, and hardware operation. Browser simulation supports demonstrations but is not a physical wristband performance test.
Evaluation	Conclusions are limited by the participants, learning content, equipment, and conditions included in testing. Broader educational benefits require further evidence.

The design takes into account that students can navigate the interface independently or with the appropriate assistance, and that learning materials have been appropriately reviewed for pedagogical accuracy. Webcam usage and record-keeping would also require the appropriate consents and accommodations.

The project does not claim to offer clinical diagnosis, to make definitive inferences about emotions or understanding through webcam observation, or to guarantee results on examinations. The chatbot analyzes the provided state information and feedback given by the learner to adjust its discourse. Scoring and indicators of progress are automated and intended to facilitate revision and recommendation, not formal assessment. The project maintains its boundaries with the above limitations while exploring the accessibility of the learning workflow.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

SignLearn AI is an integrated learning platform intended to support deaf and hard-of-hearing students studying the Sri Lankan Grade 10 and 11 ICT syllabus. The value of the system lies in the association between the lesson participation and the relevant learning support. In one environment, students can watch lessons, answer concept-based questions, request explanations, view sign demonstrations and complete practice activities.

The learning workflow commences with a syllabus-aligned lesson. While the video is being played back, attention monitoring records observations that may indicate a missed lesson interval. The knowledge graph links playback timestamps to ICT concepts and allows the selection of questions relating to the current topic, its prerequisites and related ideas. These records offer context for identifying material that may require further review.

The adaptive chatbot builds on the workflow by providing explanations and revision support linked to the learner's topic and needs. Students may request simpler explanations, examples, short notes or examination-focused guidance. Attention-based recommendations assist in directing them towards potentially missed content, and explicit feedback and question responses offer additional context for adaptation.

The sign support is a further aspect of the same learning experience. The sign-avatar component offers supported ICT terms through repayable demonstrations, and the sign-practice interface allows students to attempt relevant signs and receive feedback. The smart wristband complements the visual interface with configurable vibration alerts for selected learning events, such as prompting questions or attention reminders.

The four components share curriculum context and relevant learner records to allow support to be maintained in line with the students' lesson. For example, a potentially missed networking segment can lead to a concept question, a chatbot explanation and a relevant sign demonstration. Practice results and learning histories may then support further revision and teacher guidance.

The intended end-to-end outcome is a continuous process of learning, identifying possible difficulties, receiving suitable support and checking progress. The platform's contribution lies in how its components cooperate around the learner's needs, with the usefulness and reliability of the connections being evaluated through integrated evaluation.

3.2 High level architecture

SignLearn AI employs an application programming interface, a FastAPI server, and MongoDB database to ensure the integration of its four research components. Students interact with lessons, questions, chatbot, signers demonstrations, and practice modules through the React interface. In addition, teachers access the application to manage supported learning content and track their students' learning progress. Figure 3.1 below illustrates the main system boundaries, interfaces, and data flows.

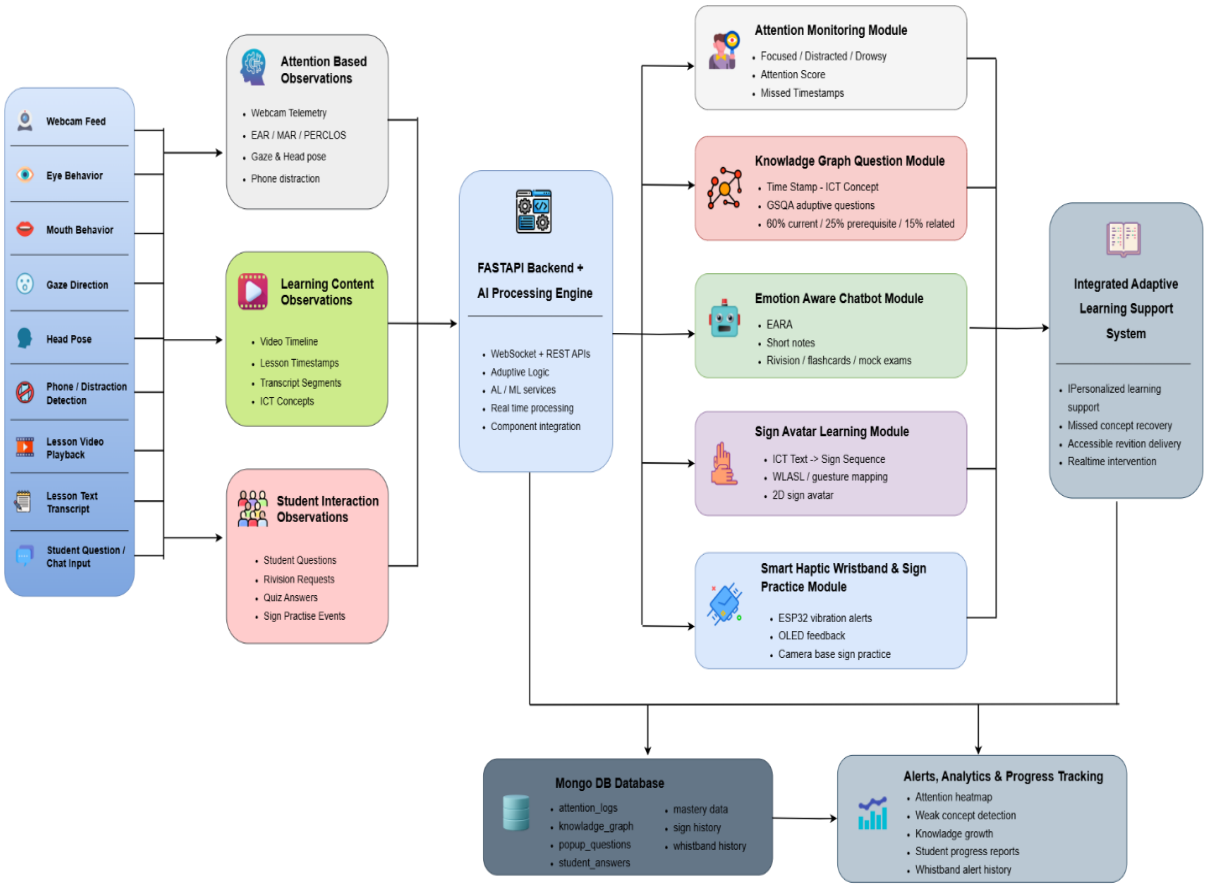


Figure 3.1: High-Level Architecture of the Integrated SignLearn AI Platform

The client browser serves the learning interface and captures webcam input for relevant activities. REST APIs provide operations to retrieve lessons, submit answers and request chatbot responses. A WebSocket connection is used for the exchange of webcam frames and attention updates during lesson playback. The backend services process these requests, and coordinate the access to shared learning records.

The first research component combines attention monitoring and questions based on a knowledge graph. Attention observations are associated with lesson timestamps, while the knowledge graph establishes lesson intervals and ICT concepts and their prerequisites. The second component provides adaptive chatbot explanations and

revision support using available curriculum and learner context. It connects to a configured language-model service and uses the local ICT dataset as a fallback if that service is unavailable, or returns an invalid response.

The third component generates sequences of sign-avatar for supported lesson terms and provides landmark data for visual replay. The fourth component supports camera-based sign practice, records learning progress and manages haptic notifications. The ESP32 wristband defines a separate physical boundary, receiving configured alerts for vibration and OLED feedback.

MongoDB stores lesson information, attention records, concept relationships question attempts, chatbot histories, sign-lecture records and wristband events. Shared identifiers for students, sessions, lessons and concepts allow these records to support interactions between components. For example, attention records can inform chatbot revision recommendations while sign demonstrations support activities in the practice interface.

The boundaries in Figure 3.1 distinguish browser interaction, backend processing, persistent storage, external services and wearable hardware. They represent the logical organization of the system, while the four research components do not need to be deployed as four separate services.

3.3 Component 1 summary

The Attention Monitoring and Knowledge Graph-Based Adaptive Question component was created by Tharaka W A K N (IT22211200). It solves the research problem of determining potentially missed lesson intervals during which a student might benefit from review, as well as the challenge of selecting relevant ICT concepts linked to the lesson.

It uses webcam frames, lesson playback timestamps, transcript segments, curriculum mappings and student question responses as inputs. The attention monitoring process uses MediaPipe-based observations, including eye closure, mouth opening, gaze direction and head pose, to capture attention indicators and potentially missed lesson intervals. The knowledge graph associates lesson timestamps with Grade 10 and 11 ICT concepts and allows question selection using the Graph-Based Question Selection Algorithm (GQSA), using weights of 60% for the current concept, 25% for prerequisite concepts and 15% for related concepts.

The component produces attention histories, timeline heatmaps, potentially missed segments, concept-based popup questions and answer records. These outputs can support chatbot revision recommendations, selected wristband alerts and teacher review. Attention observations offer supporting context, rather than direct evidence of understanding.

Its contribution to the integrated solution is linking lesson participation with curriculum-based assessment and review opportunities. Detailed detection methods, question-selection logic and evaluation evidence are presented in the owner's individual thesis.

3.4 Component 2 summary

The Emotion-Aware Adaptive Chatbot component is the work of Kothalawala L S S (IT22185884). It focuses on providing explanation and revision support that takes into account the learner's current ICT topic, questions and learning difficulties.

Its inputs consist of the student's questions, the selected Grade 10 or 11 syllabus topic, explicit learner feedback, conversation history, question responses and available attention records from Component 1. The Emotion-Aware Response Adaptation (EARA) engine processes this context to adapt explanation style and support, including simpler wording, practical analogies and examination-focused responses. The chatbot utilizes a configured language-model service, with a local ICT dataset fallback mechanism in case of its failure or unavailability.

The component generates adaptive answers, topic summaries, short notes and personalized revision recommendations. It also provides micro-challenges, flashcards, past-paper answer feedback, mock examinations and daily diagnostic quizzes. The interaction records are used to track knowledge-growth trends, repeated query alerts and teacher analytics. The learner-state indicators are used to adapt the responses but are not treated as definitive proof of emotion or understanding.

Its contribution to the integrated solution is in its ability to transform curriculum and learner context into relevant explanatory and revision support. It utilizes the attention information provided by Component 1 and connects the supported ICT keywords to the sign-avatar previews from Component 3. Its detailed adaptation rules, implementation methods and evaluation evidence are presented in the owner's individual thesis.

3.5 Component 3 summary

The Sign Avatar Lecture Generator component is created by Mendis L B L (IT22197764) to address the need for a sign demonstration that can be replayed to access the Grade 10 and 11 ICT lessons' contents.

The lessons' scripts, teacher notes and ICT-related vocabulary can be used as the component inputs. Its main pipeline processes the input text into sign glosses based on rule-based processing and displays relevant avatar-generated landmark sequences with gestures from a gesture dataset. A different WLASL pipeline uses a bidirectional LSTM to classify signs-words and provides the corresponding landmark sequences recorded for the supported vocabulary items. The classifications do not include sign generation, but only provide the movement sequences. A Sign MNIST pipeline supports the component by facilitating static alphabet recognition.

It enables the generation of signage sequences and lectures that can be saved for further use by teachers and replayed by learners. The lectures offer visual demonstrations of the selected vocabulary items alongside the written lectures, transcripts, and video content. Its limitation includes the use of only supported vocabulary items since it cannot be considered to represent the local curriculum even if the lessons included were aligned with the Grade 10 and 11 ICT lessons.

Its contribution to the overall solution involves providing shared sign demonstrations. Component 2 can offer sign previews for supported chatbot

keywords, whereas Component 4 can be used to demonstrate the signs and provide references when practicing. It is responsible for lecture and sequence generation, but not practice evaluations and haptic feedback, which belongs to Component 4. The processing details, including the preparation of the datasets and evaluation results are documented in the owner's individual thesis.

3.6 Component 4 summary

The Smart Haptic Wristband and Interactive Sign Practice component has been produced by Nawarathna I K K R B (IT19131948). It addresses the requirements of non-audio learning notifications and practice of sign vocabulary related to sign words for Grade 10 and 11 ICT topics.

Its inputs are learning events from other components, the student's notification preferences, sign demonstrations from Component 3 and webcam frames during practice. The wristband service maps learning events to configurable vibration patterns and OLED messages on an ESP32-based device. During sign practice, MediaPipe hand tracking provides landmarks which are compared to the expected gesture by how much the fingers are extended, the palm orientation and movement. Students view the reference demonstration alongside their camera feed and receive feedback.

The component produces tactile notifications, on-screen practice feedback, gesture matching scores and course progress. It maintains wristband settings and notification histories and completed practice activities contribute to badges and course completion records. Gesture matching scores indicate how well they matched the system's supported reference patterns but do not determine overall sign-language proficiency.

Its contribution to the overall solution is connecting lesson support with physical notifications and active practice. Attention events from Component 1 and appropriate learning prompts can trigger haptic alerts and Component 3 provides demonstrations for practice activities. This component manages practice evaluation, progress and feedback completing the learning sequence from viewing an explanation to attempting the associated sign. The detailed hardware design, evaluation rules and testing evidence are provided in the owner's individual thesis.

3.7 End to end user and data flow

The SignLearn AI learning process commences with a student logging in and selecting a Grade 10 or 11 ICT topic and lesson. The chosen curriculum context informs the lesson participation, concept questioning procedures, chatbot explanations and sign activities. Figure 3.2 below shows how the four research components share information in this learning process.

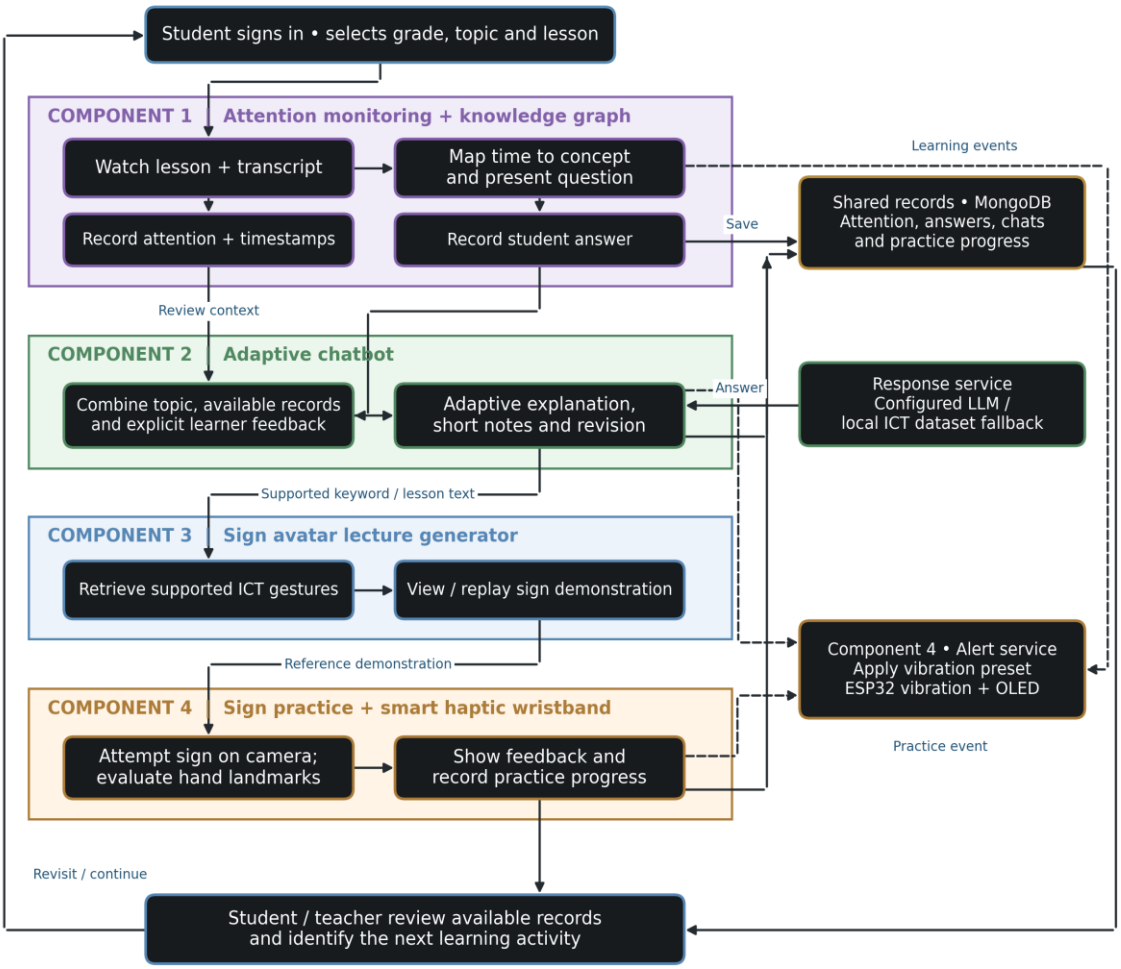


Figure 3.2: End-to-End User and Data Flow of SignLearn AI

During playback of the lesson, Component 1 offers transcript support and, if the webcam access is granted, observes attention against the timestamps of the lesson. The knowledge graph associates playback time with related ICT concepts and chooses questions that cover the current, prerequisite or related knowledge. Student responses are distinguished from the attention observations and participate in subsequent reviews.

Component 2 employs the selected topic, student questions, explicit feedback and any available learning records to offer adaptive explanations and revision. For instance, a potentially missed section of the lesson could be used to recommend a reassessment of the relevant concept. The student can request alternative wording, analogy or examination guidance. The response is generated using the configured language-model service, or defaults to a local ICT dataset, if the service becomes unavailable or malfunctions.

When a visual sign support is requested, Component 3 recalls gesture sequences for the relevant ICT vocabulary and offers replayable avatar demonstrations. The

demonstrations are accessible via the supported chatbot keywords and sign lectures. Component 4 references the demonstrations during camera-based practice, assesses the hand landmarks and records the feedback and progress. Its wristband service converts relevant learning events into configured vibration patterns and OLED messages. The physical notification is delivered if a device connection is active.

REST APIs facilitate exchanges between the frontend and backend, and a WebSocket connection streams the live attention frames and status updates. Student, session, lesson and topic identifiers associate relevant records in MongoDB. The lesson videos, gesture datasets and model assets are stored separately on the server. The available records offer student review and teacher guidance, identifying appropriate follow-up activities.

The students can re-visit or access the activities as per their needs rather than completing the whole sequence in a given order. Attention observations suggest potential reviews, which are considered together with the learner feedback and question responses instead of being perceived as proof of understanding.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

SignLearn AI makes use of one shared React frontend, one FastAPI backend and one MongoDB database across all four research components. Each part is responsible for its own processes and data records. In turn, defined interfaces let the connected learning activities take place. Student identity, lesson timestamps and curriculum mappings provide the required context to exchange the needed information between the components.

The integration sequence follows the activities' dependencies within the learning process. Authentication, lesson access and curriculum context serve as the basis upon which attention monitoring and knowledge graph questions build. The chatbot makes use of available attention records and topic mappings to offer revision suggestions. Sign demonstrations are linked to supported chatbot keywords and the practice interface, while the wristband service manages the relevant notification events. Table 4.1 provides an overview of the described integration stages and the verification focus.

Table 4.1: Integration Sequence and Verification Focus

Stage	Integration focus	Verification focus
Shared foundation	Authentication, lesson access and curriculum context	Consistent student, lesson and topic references
Lesson support	Attention observations, playback timestamps and concept questions	Correct association between recorded events and lesson intervals
Adaptive revision	Attention records and topic context supplied to the chatbot	Recommendations refer to the appropriate lesson concepts
Sign support and practice	Avatar demonstrations accessed through chat and practice activities	Supported vocabulary retrieves the intended demonstration
Haptic feedback	Learning events passed to the wristband service	Correct preset selection, event recording and physical delivery where connected

Interface ownership is defined by the research division: Tharaka W A K N owns attention and knowledge graph interfaces, Kothalawala L S S owns chatbot and revision interfaces, Mendis LBL owns sign-generation and landmark-sequence interfaces and Nawarathna I.K.K.R.B owns practice-progress and wristband interfaces. For each connection, the supplying component defines the available output and the consuming component processes that output within its learning

workflow. Therefore, changes to identifiers, request fields or response formats necessitate co-ordination between the relevant owners.

The integration environment comprises the frontend development application, the backend services, MongoDB and the required datasets and model assets. Configured language-model services and the ESP32 wristband introduce external dependencies. Browser-based wristband simulation enables interface checks, whereas physical hardware is required to verify vibration and OLED delivery.

Integration risk management focuses on consistent data mappings and explicit resolution of unavailable dependencies. Request and response models define the exchanged information between APIs and the chatbot's local dataset fallback reduces its dependence on its configured language-model service. Missing attention records must remain distinguishable from recorded low attention and notification logs must remain distinguishable from confirmed device delivery. Chapter 6 presents the integration tests and evidence used to evaluate these behaviors.

4.2 Interfaces data contracts and shared models

The four research components are integrated with each other through the use of REST APIs, a WebSocket connection for the exchange of attention-related information, and common database records. The bodies of REST requests and responses are encoded in JSON, with the exceptions of operations that require file uploads or provide downloadable resources. The backend uses Pydantic models to define the structure of request and response bodies. Table 4.2 below contains a brief overview of the main API endpoints.

Table 4.2: Main Interfaces and Data Exchanged Between Components

Interface	Main data exchanged	Responsible component
POST /api/videos/upload and GET /api/transcripts/{video_id}	Lesson uploads, video references and timed transcript segments	Component 1
WS /ws/attention/{session_id}	Encoded webcam frames and attention-status updates associated with a session	Component 1
GET /api/knowledge-graph/lesson/{lesson_id}/question	Playback time supplied as a query parameter; selected concept and question returned	Component 1
POST /api/knowledge-graph/answer	Student question attempt and answer-evaluation result	Component 1
GET /api/chatbot/attention-	Revision suggestions derived from available attention	Component 2, using

recommendations/{student_id}	records and lesson mappings	Component 1 records
POST /api/chatbot/chat	Student question, curriculum and learner context; adaptive answer and response-source information	Component 2
POST /api/sign-avatar/generate	Input text and the resulting supported gloss and gesture sequence	Component 3
GET /api/signs/landmark-sequence/{gloss_word}	Landmark sequence for a supported sign word	Component 3
POST /api/sign-course/progress	Student practice outcome and course-progress update	Component 4
POST /api/wristband/notify	Learning notification details used to resolve a vibration preset and record the event	Component 4

Shared identifiers link records in these interfaces. Student references link learning activity to a user, and session identifiers group attention observations. Video and lesson references link timestamps to transcripts and concept windows. Concept, topic and sign references give the curriculum links used for revision and practice. Where routes use different names, such as user_id and student_id, their relationship must be explicitly mapped. Similarly, video_id and lesson_id should only be treated as equivalent where the implementation establishes that relationship.

The principal logical data models cover lesson context, attention observations, question attempts, chatbot interactions, sign sequences, practice progress and wristband events. Each combines the relevant identifiers with its activity-specific information. For example, a question attempt associates an answer with a student and question, and a practice record associate an outcome with a student and sign. Chatbot responses additionally expose sourceType, fallbackReason and confidence to describe the response source and fallback status. These fields do not independently establish answer correctness.

Validation responsibilities are divided between the interface and service layers. Request models check declared field types and required values. Service-level checks are needed for referenced records, valid question options, supported vocabulary and meaningful timestamps. Frontend checks provide immediate feedback, while backend checks determine whether a request can be processed. Camera frames sent through the attention WebSocket require validation within the WebSocket handler rather than relying solely on REST request models.

Error handling belongs first to the component providing the interface. It must distinguish invalid input, missing resources and processing failures so that the consuming interface can display an appropriate message. The chatbot handles configured language-model failures through its local dataset fallback. Unavailable

gesture data must be distinguishable from a successfully generated sequence, and a recorded wristband notification must remain distinguishable from confirmed physical delivery.

The documented routes use the /api prefix without an explicit version segment. No common schema-version field or uniform error-response format is established in the supplied documentation. Consequently, changes to field names, required values or response structures require coordination between component owners and corresponding updates to consumers and integration tests. Backend request and response models provide the reference for interface field definitions. Changes to these models require corresponding updates to consuming components and integration checks.

4.3 Shared nonfunctional requirements

The integrated platform should offer an easily available and reliable learning environment of the four components. Its quality will depend on how lessons, explanations, sign activities and notifications are interrelated. Table 4.3 shows the shared nonfunctional requirements and their evidence, which are criteria for the evaluation rather than statements that all have been met.

Table 4.3: Shared Nonfunctional Requirements and Evaluation Evidence

Requirement	Application to SignLearn AI	Evaluation evidence
Performance	Video playback and navigation should remain responsive during attention monitoring. Chatbot responses, sign playback and notifications should arrive within limits appropriate to their learning activity.	Response times, attention-update delays, playback interruptions and notification latency under stated test conditions
Usability	Students should be able to select topics, request explanations, replay demonstrations and complete practice with clear instructions and consistent navigation.	Task-completion rates, observed difficulties and student or teacher feedback
Security	Authentication and server-side authorization should restrict access to student records, teacher functions and administrative operations. Uploaded content and API inputs require validation, and service credentials must remain outside frontend code.	Access-control tests, invalid-input tests and configuration review
Privacy	Webcam processing and learning-	Review of consent

	record collection should have a clearly explained purpose. Collection, retention and access should be limited to what the learning and evaluation activities require.	information, data flows, retention arrangements and access permissions
Reliability	Dependency failures should produce understandable feedback. The chatbot should provide its documented fallback, while missing camera data or disconnected hardware should be represented accurately.	Tests covering service timeout, unavailable camera input, missing resources and wristband disconnection
Maintainability	Component responsibilities, request models and shared mappings should remain clearly organized and documented so changes can be coordinated across owners.	Code review, API documentation and regression checks for affected interfaces
Scalability	Increasing concurrent lesson sessions, webcam traffic and stored histories should be assessed against available processing and storage capacity.	Load-test results, resource utilization and response-time changes as concurrent usage increases
Accessibility	Essential content and feedback should be available without relying on sound alone. Text, transcripts, sign demonstrations and tactile notifications should complement clear navigation and readable interfaces.	Keyboard checks, contrast and text-resizing checks, transcript review and task-based accessibility feedback

Accessibility assessment should use the Web Content Accessibility Guidelines (WCAG) as a reference about interface requirements that would be applicable, including keyboard operation, visible focus, text contrast and alternatives for relevant media. Providing sign demonstrations or a wristband alone is not sufficient to confirm accessibility compliance. The report should indicate which WCAG version and criteria has been assessed and differentiate checks from formal conformance claim.

Privacy requirements are particularly relevant because attention observations and learning histories relate to individual students. The system should inform when the camera is active and differentiate processing of transient frames from any retained images or derived records. Where participants are minors, the consent and assent arrangements should relate to the institution's approved research procedures.

Across the requirements, reporting should be accurate and traceable: missing observations should not be interpreted as poor attention, gesture-matching scores should not be presented as general sign-language proficiency and recorded notifications should not be treated as confirmed physical delivery. Evaluation against

the requirements considers the outcomes measured, test conditions and remaining limitations.

4.4 Integrated workflow and exception handling

Normal operation begins when a student signs in and picks an ICT lesson. Component 1 fetches the lesson and transcript, correlates attention observations with playback timestamps, and poses knowledge graph questions. Recorded observations and available mapping of concepts inform revision recommendations in Component 2, where the student may request explanations or attempt revision activities. Component 3 offers demonstrations for supported sign vocabulary, and Component 4 supports camera-based practice and relevant wristband notifications.

These interactions depend on different resources, including webcam access, stored lesson mappings, configured language-model service, gesture assets and connected hardware. Exception handling thus needs to distinguish between unavailable feature and valid learning outcome. For example, a missing camera frame can not establish distraction, and an unsuccessful progress update can not establish that a practice attempt has been saved.

Table 4.4 lists the main failure paths and recovery requirements. The language-model fallback is explicitly described in the implementation documentation, the remaining recovery behaviors require verification through integration testing.

Table 4.4: Integrated Failure Paths and Recovery Behavior

Failure or exception	Affected workflow	Recovery behaviour or fallback
Webcam permission denied or camera unavailable	Attention monitoring and camera-based sign practice	Explain that camera-dependent processing is unavailable. Keep accessible lesson and revision activities available, and allow the student to retry camera access.
Attention WebSocket disconnects	Live attention updates	Indicate the connection interruption and attempt or offer reconnection. Treat the interrupted interval as missing data rather than inferred distraction.
Lesson media or transcript cannot be retrieved	Lesson playback and text support	Identify the unavailable resource and offer retry or another lesson. Avoid presenting a failed

		retrieval as empty lesson content.
No valid timeline or concept mapping is available	Concept questions and targeted revision	Avoid assigning an unsupported concept. Permit manual topic selection or general topic revision where available.
Configured language-model service fails or returns an invalid response	Adaptive chatbot	Use the local O/L ICT dataset fallback and return the documented response-source and fallback information.
Requested sign has no supported gesture sequence	Avatar demonstration and related practice	Indicate that the demonstration is unavailable and offer supported vocabulary. Do not substitute an unrelated gesture as a correct sign.
Hand landmarks cannot be evaluated reliably	Sign practice	Request improved hand visibility, positioning or lighting. Do not interpret an unevaluable attempt as a confirmed incorrect sign.
Wristband is disconnected or delivery is unconfirmed	Physical notifications	Retain visible feedback and distinguish the recorded notification event from confirmed hardware delivery. Offer reconnection or a device test.
Answer or progress record cannot be saved	Learning history and progress tracking	Report the unsuccessful save and offer a controlled retry. Prevent repeated submissions from creating duplicate records.

Responsibility for detecting an exception is a feature of the component processing the request. The frontend is then used to translate the result into understandable feedback, and provide any available recovery action. Errors passed between components should retain sufficient context to identify the affected activity while avoiding exposure of credentials or other student information.

Recovery should also preserve record consistency - reconnection, retries and repeated button presses must not silently duplicate question attempts, practice

completions or notification events. Integration tests should verify both the visible user response, and resulting database state, including whether interrupted activities can resume without producing misleading learning records.

4.5 Team contribution and dependency matrix

The project distributes responsibility among four research components, each of which is responsible for the associated research, development and component-level evaluation. Integration links these contributions through shared curriculum context, learning records and interfaces. Table 4.5 shows the members' responsibilities and dependencies. The evidence column indicates what records are needed to provide the support of each contribution; it does not imply that all tests are finished.

Table 4.5: Team Contributions, Dependencies and Supporting Evidence

Member and component	**Research and development responsibilities**	**Integration work and dependencies**	**Evidence Required for Verification**
Tharaka W A K N - IT22211200 Component 1: Attention monitoring and knowledge graph questions	Investigate attention indicators and concept-based question selection. Develop webcam processing, lesson timeline mapping, question selection, answer recording and attention reports.	Provide attention and lesson context for Component 2 revision recommendations and relevant events for Component 4 notifications. Maintain mappings between lesson intervals and curriculum concepts.	Annotated attention comparisons, timestamp-to-concept checks, question-selection tests, answer-persistence checks and cross-component request logs
Kothalawala L S S - IT22185884 Component 2: Emotion-aware adaptive chatbot	Investigate response adaptation and curriculum-based revision. Develop chatbot responses, local fallback, short notes, revision activities, learning analytics and repeated-query alerts.	Consume available Component 1 attention records and topic mappings. Connect supported chatbot keywords to Component 3 demonstrations and relevant notifications to Component 4.	Response-review records, adaptation scenarios, fallback tests, recommendation-mapping checks and analytics consistency checks
Mendis L B L - IT22197764	Investigate supported text-to-	Supply supported sign demonstrations	Gesture-retrieval checks, playback

Component 3: Sign avatar lecture generator	sign representation and landmark-based playback. Develop gesture lookup, sign lectures, saved sequences and the separate WLASL and Sign MNIST pipelines.	to Component 2 previews and Component 4 practice. Coordinate vocabulary references and sequence formats with consuming interfaces.	and saved-lecture tests, model-evaluation records and demonstration-to-practice mapping checks
Nawarathna I K K R B - IT19131948 Component 4: Smart haptic wristband and sign practice	Investigate tactile learning notifications and camera-based practice feedback. Develop ESP32 firmware, vibration presets, device interfaces, gesture evaluation and course-progress tracking.	Receive relevant learning events from Components 1 and 2 and use Component 3 demonstrations during practice. Coordinate notification formats and progress records.	Physical vibration and OLED tests, delivery-latency records, disconnection tests, practice-feedback comparisons and progress-persistence checks

Shared outputs are the integrated frontend and backend, curriculum mappings, interface documentation, system diagrams, end-to-end test scenarios and the common report. These output items require contributions from both supplying and consuming component owners. For instance, the avatar owner provides a supported sequence whereas the practice owner makes sure the corresponding activity gets and shows it properly.

The responsibility matrix separates individual from shared component ownership. The interface always has a supplying and a consuming component, so integration contributions cut across components. Individual theses describe component-level contributions whereas the common report covers shared interfaces and combined system behavior.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

SignLearn AI uses shared web infrastructure to realize the four research components within a common project structure and allow each member's functionality to interact via common interfaces. Table 5.1 summarizes the main technologies and reasons for their use.

Table 5.1: Common Technology Stack and Infrastructure

Layer or resource	Technology	Role and reason for selection
Frontend	React, Vite and Tailwind CSS	Provide reusable learning interfaces and consistent styling across lessons, chatbot interactions, sign demonstrations and practice activities. Vite supports frontend development and build preparation.
Backend	Python and FastAPI	Provide REST endpoints and WebSocket support while allowing direct integration with the project's Python-based vision and machine-learning utilities.
Request and response models	Pydantic	Define structured API models and validate declared field types and required inputs.
Database	MongoDB	Stores lesson metadata, concept relationships, attention observations, question attempts, chatbot histories and component-specific records. Document structures accommodate the different record types produced by the platform.
Live communication	WebSocket	Carries webcam frames and attention updates during lesson playback without requiring a separate HTTP request for each exchange.
Vision processing	MediaPipe and OpenCV	Support landmark extraction and video-processing tasks used by attention monitoring, sign-related pipelines and practice activities.

Chatbot services	Configured language-model service and local O/L ICT datasets	Support adaptive responses, with local dataset fallback when the configured service fails or is unavailable.
Sign-model infrastructure	TensorFlow/Keras and scikit-learn	Support the separate WLASL LSTM and Sign MNIST MLP pipelines. These models are distinct from the primary rule-based avatar pipeline.
Wearable infrastructure	ESP32, SSD1306 OLED and haptic vibration motor	Provide physical notifications and short visual messages for relevant learning events.
File storage	Backend upload and dataset directories	Maintain lesson media, curriculum resources, gesture assets and trained model files separately from database records.

The project has common frontend/ and backend/ folders with a specific feature modules. Even though the code is organized into five modules, the research work consists of four parts: attention monitoring and knowledge graph questions are related to the same member. The architecture allows implementing several components with a clear separation of responsibilities but as a part of the same application.

The development environment combines the frontend development server, FastAPI app and MongoDB instance with datasets and model assets needed for the project. The language model configuration is defined with environment variables for the backend part, including provider, model and timeout. API keys for the backend part should be placed in the configuration but kept out of the shared code. For the physical device evaluation, an additional ESP32 dev board is needed with Bluetooth connection.

The infrastructure supports the project's requirements without the need to deploy each research component as a separate service. REST API is used for regular requests and WebSockets for real-time attention data transmission. Larger binary assets and models require file storage that is available separately. Hosting, deployment tools and environment-specific variables should be based on the actual code structure: the infrastructure code defines the tech stack but not the specific cloud provider.

5.2 Development and integration process

The evolution of SignLearn AI moved from the proposal-stage design towards an integrated implementation, with the changes mirrored in the progress presentations and component documentation updates. The four members maintained the responsibility for their respective areas of research while adjusting their

responsibilities according to the Grade 10 and 11 ICT course. The project's primary development goals included establishing attention records' connection to revision support, enabling the availability of sign demonstrations across learning activities and implementing practice feedback and wristband notifications.

The work planning emphasizes the components' tasks and their interdependence, meaning that a feature utilizing another component's output requires negotiating the supplying interface before finalizing the entire process. In other words, attention-driven revision relies on the availability of timestamps and concept association within lessons, while sign practice utilizes the existing vocabulary and demonstrations' availability. Documentation, in this case, supplies the necessary details regarding the interconnections but does not standardize the process, duration of sprints, or overall Scrum methodology.

The API catalogue and request/response models offer the framework for coordination, as the owners of particular components can work on their processes independently while utilizing the information regarding identifiers, provided input and output, and required parameters from other modules. Nevertheless, the existing records do not clarify whether the interface agreements were reached before individual implementations, and therefore the interface-first development approach cannot be confirmed based on the architecture alone.

Table 5.2: Development and Integration Activities and Supporting Records

Activity	**Application to the project**	**Supporting record**
Planning and task allocation	Divide component work and priorities tasks.	Dated task plans, sprint records or meeting notes
Interface coordination	Agree on curriculum references, request fields, response structures and failure behavior	API documentation, model changes and interface decisions
Version control	Track component changes and coordinate updates to shared files	Commit history, branches and merge records
Review	Check implementation changes and their effects on dependent components	Review comments, pull requests or recorded walkthroughs
Issue management	Assign integration defects to the relevant owners and track their resolution	Issue records, reproduction steps and retest outcomes
Continuous integration	Automate applicable build and test checks where a pipeline is configured	Pipeline configuration and execution logs

Release preparation	Assemble a consistent application, configuration and required assets for evaluation	Release checklist, revision identifier and test report

Integration points cover both complete interactions and individual features. Considered scenarios include lesson retrieval, answer recording, revision suggestion generation, supported sign demo open and practice progress update. Failures should be tracked to the involved interface, fixed by its owner and retested with the consuming component.

A repeatable evaluation release should include a recorded code revision, datasets and model references, configuration instructions and known limitations. The provided materials do not indicate a branching strategy, automated CI pipeline or release schedule. Assertions about these practices should therefore be justified by referring to the team's development records, with the supporting evidence given in Section 5.4.

5.3 Deployment and operational configuration

SignLearn AI consists of a React frontend, FastAPI backend, MongoDB database and associated lesson, dataset and model files. The ESP32 wristband represents a connected hardware component. While the provided implementation documentation describes the application architecture, it does not define a production hosting configuration or deployment pipeline. This section will therefore detail the deployment requirements and operational considerations that must be validated for the evaluation environment.

The frontend is implemented using the project's Vite configuration and is configured to consume the backend REST and WebSocket endpoints. Deployment of the backend requires the Python dependencies, database connection parameters and file paths used by each component. Curriculum datasets and lesson timeline mappings must be loaded prior to question and revision workflow execution. Gesture assets are required to enable avatar playback, while trained model files are needed for the separate classification pipelines that utilize them.

Table 5.3: Deployment Resources and Configuration Requirements

Resource	Configuration and operational requirements
Frontend	Backend API address, attention WebSocket address and browser permissions for camera-based activities
FastAPI backend	Python dependencies, database connection, permitted frontend origins and upload, dataset and model paths
MongoDB	Connection settings, required seed records and appropriate access

	restrictions
Chatbot service	LLM_PROVIDER, LLM_API_KEY, LLM_MODEL and LLM_TIMEOUT_MS, together with accessible local fallback datasets
Sign services	Gesture dictionary and landmark assets; trained model files for the enabled WLASL and Sign MNIST functions
Wristband	Uploaded firmware, functioning display and vibration hardware, and the configured communication connection

Configuration containing credentials should be kept external to the frontend and version controlled. A remotely hosted deployment requires browser security mechanisms to enable camera access and secure WebSocket connections. Pre-evaluation operational checks should include authentication, lesson retrieval, database access, chatbot fallback, sign playback and wristband connectivity.

Monitoring should differentiate between availability of the application and successful operation of learning activities. Relevant considerations include API failures, WebSocket interruptions, delayed responses, language-model fallbacks and database write failures. Wristband event records should differentiate between notification requests and physical delivery when acknowledgements are available. The supplied documentation contains histories and service-status endpoints but does not provide a monitoring or alerting platform.

Backup coverage must include MongoDB records and the file assets required to recreate the system, including uploaded lessons, curriculum datasets, gesture sequences and model files. Sensitive configuration requires protected storage. A backup schedule, retention period and recovery owner should be recorded for the actual deployment; these are not specified in the available materials.

Restoration requires database and file recovery, configuration reinstatement and application service start. Verification should confirm existing lessons, histories and saved sign lectures are accessible and new records can be created. A successful backup does not demonstrate recoverability, so results should be documented before claiming the deployment has a tested recovery process.

5.4 Evidence of team execution

The team’s contributions are captured in the proposal documents pertaining to the project, progress presentations and individual components’ documentation. The latter two showcase how the work on the project has been distributed amongst the team and how the initially proposed idea was shaped into the finally executed one. In particular, the descriptions of the progress made at different stages allow for understanding how the distribution of responsibilities for the components differed from the initially proposed plan.

In particular, the four members have collaborated on the aspects of attention monitoring and knowledge graph questions, adaptive chatbot support, sign-avatar generation, and wristband-assisted sign practice. For each of the components, the documentation highlights the interfaces, resources, and tests utilized. Overall, the papers and presentations offer the basis for describing the individual contributions to the project and, therefore, the scope of work covered by each team member. However, the documentation might need to be supplemented by the dated records of developments and tests to showcase the progress made on each component and its successful integration into the overall solution.

Table 5.4: Evidence Sources and Their Contribution to Execution Traceability

Evidence source	**Contribution to the report**
Proposal documents and progress presentations	Show the original responsibilities, reported progress and changes in component scope
Component and API documentation	Describe implemented functions, interface responsibilities and cross-component dependencies
MS Planner task records	Support task ownership, planned milestones and recorded completion status
Microsoft Teams discussions	Support coordination decisions, clarification of requirements and resolution of integration questions
Git history	Links implementation changes to contributors and records changes to shared interfaces
Meeting records	Explain decisions, assigned actions and follow-up responsibilities
Research diaries	Record individual investigations, development difficulties, experiments and reflections
Integration test records	Connect completed changes to observed system behavior and identified defects

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The evaluation investigates whether the four components of SignLearn AI are integrated correctly and offer appropriate, accessible support for Grade 10 and 11 ICT learning. The three questions posed are: does the information flow correctly between the components, does it provide appropriate support tools for the learners, and can failures be recovered from without generating misleading information or records? Table 6.1 shows the criteria proposed; it does not show the results of tests carried out.

Table 6.1: Evaluation Areas and Proposed Success Criteria

Evaluation area	Metrics and reference	Success criterion
Integrated workflows	Scenario completion and record consistency against predefined expected outcomes	Critical workflows complete with correct student, lesson and topic associations
Revision and explanations	Mapping accuracy and teacher-rated correctness and relevance, compared with annotated timelines and standard explanations	Recommendations match the lesson context, and reviewed explanations contain no material factual errors
Sign demonstrations and practice	Gesture retrieval, playback and agreement with expert-reviewed references	Supported vocabulary displays the intended demonstration; unevaluable attempts are identified clearly
Wristband feedback	Pattern correctness, physical delivery and latency against configured presets	Connected-device cases produce the expected alert without falsely confirming disconnected delivery
Usability and accessibility	Task completion, assistance and observed barriers compared with lesson-and-transcript tasks	Essential activities have no unresolved blocking usability or accessibility issues
Reliability and security	Failure recovery, record integrity and unauthorized-access tests	Critical failures preserve consistent records, and tested unauthorized requests do not expose protected data

Evaluation materials consist of selected ICT lessons, transcripts, concept questions, chatbot prompts, supported gestures and notification scenarios. The intended participants are deaf and hard-of-hearing learners, ICT teachers and sign-language reviewers. Actual sample sizes and recruitment criteria need to be documented in conjunction with the results; developer evaluations are kept separate from stakeholder reviews.

Analyses will incorporate completion rates, error numbers, response latency, reviewer ratings and repeated feedback themes. Numerical performance criteria need to be established in advance of formal trials. Technical success and positive usability reports are distinguished from gains in comprehension, which require direct measures and a suitable control condition for comparison.

6.2 Integration and end to end testing

Integration testing covers whether the four components communicate properly with one another and end-to-end testing follows the whole of student and teacher activities through the platform. The tests cover normal operation, missing dependencies and consistency of saved records. Table 6.2 defines the relevant scenarios; execution outcomes require supporting test records.

Table 6.2: Integration and End-to-End Test Coverage

Test category	Main scenario	Expected outcome
Interface	Exchange lesson timestamps, concept references, chatbot requests and sign sequences	Valid requests return the expected structures; invalid inputs receive understandable errors
System and end-to-end	Open a lesson, answer a question, request revision, view a sign and complete practice	Relevant curriculum context is preserved, and records are associated with the correct student
Regression	Repeat core workflows after changes to shared interfaces	Previously working functions remain operational without duplicate or inconsistent records
Security	Attempt access to another student's history or restricted teacher functions	Unauthorized access is rejected without exposing protected information
Performance	Run lesson playback alongside attention monitoring and concurrent requests	Response times, update delays and resource use are measured against the criteria established in Section 6.1
Reliability	Interrupt the WebSocket, disable the language-model service or disconnect the	Unavailable functions are identified accurately, and documented fallbacks operate as

	wristband	expected
Usability	Navigate lessons, request explanations and use sign replay and practice controls	Task completion, assistance required and accessibility barriers are recorded
User acceptance	Ask representative learners and teachers to complete realistic learning tasks	Feedback establishes whether the tested workflows meet their needs and identifies remaining issues

Each executed case is documented with a test ID, environment, input, expected result, observed result and outcome. We cover both the visible interface and the corresponding database record. Wristband tests must observe the physical feedback, as the mere presence on a notification log does not guarantee it

s successful delivery.

Failed cases must be associated to a defect and re-executed following the correction, including inspection of the affected workflows. Developer and stakeholder acceptance tests are reported separately. Pass rates, performance results and acceptance findings only consider executed cases with recorded evidence.

6.3 Overall results

The implementation documentation on hand refers to an integrated platform, which links lesson participation and the subsequent concept questions, adaptive explanations, sign demonstrations and practice with haptic feedback. Table 6.3 summarizes the documented outputs and differentiates them from outcomes in need of measured evaluation evidence.

Table 6.3: Documented Group-Level Outputs and Evaluation Status

Area	**Documented system output**	**Evaluation status**
Lesson support	Attention records, concept-based questions and lesson review information	Mapping accuracy and end-to-end reliability require recorded test results
Adaptive revision	Topic-based explanations, revision recommendations and local dataset fallback	Response quality and fallback reliability require reviewed outputs and execution records
Sign support	Supported avatar demonstrations and camera-based practice feedback	Demonstration suitability and practice-evaluation accuracy require independent assessment

Haptic feedback	Configurable wristband notifications and event histories	Physical delivery rate and latency require device measurements
Shared learning records	Student histories, progress information and teacher analytics	Cross-component consistency requires comparison with underlying records

These outputs represent the scope of the documented implementation. However, the evidence available for this report does not yet establish an overall test pass rate, distribution of response times or stakeholder acceptance result. No numeric improvement in attention, understanding or sign proficiency can be concluded from the feature descriptions alone.

The individual theses provide the appropriate locus for conducting detailed component experiments, such as attention detection, question selection, chatbot response evaluation, sign classification and practice feedback. Component-level measures must be differentiated from group-level results: classifier accuracy, for instance, does not establish the quality of avatar demonstrations or the effectiveness of the entire learning workflow.

The current group-level conclusion is that the documented design integrates the four research contributions around common ICT learning activities. Claims about the reliability, accessibility and educational effectiveness of such connections remain contingent on the integrated testing and stakeholder evidence described in Sections 6.1 and 6.2.

6.4 Stakeholder and user validation

Stakeholder validation confirms if SignLearn AI delivers comprehensible, pertinent and accessible support throughout the learning process. The main target users are deaf and hearing-impaired students that study Grade 10 and 11 ICT. Teachers and sign-language reviewers evaluate the relevance of the tool from different angles, addressing the accuracy of the signs, learning support, and the appropriateness of the given examples.

The documented development iterations improved syllabus relevancy, the clarity of the avatar's hand and finger movements, and the signs' topic-based organization. Table 6.4 cross-references the enhancements and illustrates the remaining validation criteria.

Table 6.4: Stakeholder Needs, Documented Refinements and Validation Areas

Stakeholder perspective	Documented feature or refinement	Remaining validation area
Learners accessing lessons	Syllabus-organized videos, transcripts and topic-linked revision	Whether learners can locate and revisit relevant material independently
Learners requesting explanations	Curriculum selection, simpler explanations, analogies and examination-focused support	Whether explanations are understandable and appropriate to learners' needs
Learners viewing and practicing signs	Clearer avatar movements and practice activities organized around ICT vocabulary	Whether demonstrations are recognizable and practice instructions are clear
Learners using tactile notifications	Configurable vibration patterns and a wristband test function	Whether patterns are distinguishable, comfortable and useful during lessons
ICT teachers and special education instructors	Learning histories, repeated-query alerts and attention summaries	Whether reports help identify difficulties without overstating understanding
Sign-language reviewers	Demonstrations for supported ICT terms	Whether the signs and movement sequences are suitable for the intended Sri Lankan learner community

The refinements aim to clarify lesson access, explanations and sign practice, and to align them more closely with the ICT curriculum. They enable a cohesive learning experience in which students can revisit content, ask for guidance and practice supported vocabulary.

Stakeholder validation is required to confirm if learners will be able to work through these activities unaided, interpret the automated feedback and use the wristband comfortably. Teacher and sign-language reviews will establish the value of learning reports and the appropriateness of demonstrations for Sri Lankan learners.

6.5 Achievement of common objectives

The common objectives are compared with the documented performance and available evaluation information. Table 6.5 contrasts the provision of relevant functions with verification of their accuracy, usefulness and educational value.

'Partly achieved' is used to indicate that supporting functionality is documented, but that the full objective has not been demonstrated using sufficient evidence.

Table 6.5: Achievement of Common Objectives

Objective	**Available evidence**	**Achievement and justification**
GO1: Establish shared Grade 10 and 11 ICT curriculum context	Syllabus-organized lessons, chatbot topics, knowledge graph concepts and ICT sign activities	Partly achieved. Curriculum organisation is documented, but coverage and consistency across all components require verified mapping checks.
GO2: Connect potentially missed lesson intervals to relevant review	Timestamped attention records, lesson concept mappings and chatbot revision recommendations	Partly achieved. The supporting workflow is documented; mapping accuracy and recommendation relevance require evaluation results.
GO3: Provide adaptive explanations and revision support	EARA response adaptation, explanation modes, short notes, micro-challenges and local fallback	Partly achieved. The functions address the objective, but their correctness and suitability require structured response reviews and learner evaluation.
GO4: Connect sign demonstrations with guided practice	Avatar sequences, supported vocabulary, camera-based practice and progress records	Partly achieved. Demonstration and practice functions are documented; linguistic suitability and practice-feedback accuracy remain unverified.
GO5: Deliver appropriate tactile learning notifications	ESP32 firmware, vibration presets, notification interfaces and event histories	Partly achieved. Notification functionality is documented, but confirmed physical delivery, latency and pattern recognizability require test evidence.
GO6: Evaluate the complete platform’s usability, accessibility and reliability	Evaluation framework, test scenarios and defined quality requirements	Not yet demonstrated. Planning establishes evaluation coverage, but completed system tests

		and stakeholder findings are needed to establish achievement.

The documented implementation demonstrates progress toward the common goal of linking accessible ICT content with relevant learning support. However, full attainment requires demonstrating that these functions work correctly together and cater to the needs of the users. Thus, the present evaluation acknowledges the implementation's progress while withholding endorsement of enhanced understanding, accessibility and user acceptance until the completion of the evaluation.

6.6 Integrated limitations and threats

SignLearn AI includes a variety of learning support functions, but the quality of both individual components as well as the interconnections determines the success of the solution. A wrong lesson to concept mapping can lead to wrong question selection as well as revisions suggested. Similarly, a wrong sign demonstration could affect the choice of reference to be followed during practice. Table 6.6 provides the major limitations as well as threats to validity.

Table 6.6: Integrated Limitations and Threats to Validity

Area	Limitation or threat	Implication
Attention interpretation	Lighting, camera position, occlusion and natural movements can affect webcam observations. Looking away may also form part of normal learning activity.	Attention indicators cannot independently establish distraction or understanding.
Cross-component mappings	Lesson timestamps, topic identifiers and supported vocabulary must remain consistent across services.	Incorrect or missing mappings can produce irrelevant questions, recommendations or demonstrations.
Chatbot responses	Generated explanations may contain errors, while the local fallback is limited by dataset coverage.	Response availability does not guarantee correctness or equivalent quality across response sources.
Sign support and practice	Supported vocabulary is limited, and WLASL-based resources do not establish suitability for Sri Lankan sign-language users. Landmark matching captures only	Classification and matching scores cannot establish linguistic accuracy or general signing proficiency.

	selected aspects of signing.	
Hardware and runtime dependencies	Camera access, processing capacity, network conditions and wristband connectivity affect operation.	Delays or interruptions may disrupt monitoring and notifications; logged events do not guarantee physical delivery.
Evaluation validity	Developer-selected tasks and subjective reviews may favour familiar scenarios. Repeated exposure can influence task performance.	Independent assessment and controlled comparisons are needed to attribute benefits to the system.
Generalizability	The platform targets selected Grade 10 and 11 ICT content and supported learning activities.	Findings cannot automatically extend to other subjects, age groups or communication preferences.

System-level performance must be evaluated separately from component experiments. The overall model does not reflect successful integration, and heightened activity or attention scores do not necessarily indicate learning improvements. Without stakeholder assessment, the platform's applicability is indeterminable.

Integration risks such as disrupted WebSocket sessions, inconsistent identifiers, duplicated records during retries and indeterminate device-delivery status require end-to-end fault testing and record inspections. Future evaluation should prioritize these dependencies with independent curriculum and sign-language reviews specifying participants, tasks and environmental conditions.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

SignLearn AI's value proposition focuses on connecting accessible ICT lesson content with relevant revision, sign demonstrations and practice within one learning platform. The target users are deaf and hard-of-hearing students studying the Sri Lankan Grade 10 and 11 ICT syllabus. The shared context of lessons and topics enables students to switch between the different types of activities without having to constantly reorientate themselves as to what they are learning.

The platform combines transcripts and written explanations with supported sign demonstrations and optional tactile notifications. Teachers have access to learning records which can be used to provide follow-up guidance. Table 7.1 shows the intended value for users, potential customers and partners.

Table 7.1: Beneficiaries and Intended Value of SignLearn AI

Beneficiary	Intended value
Deaf and hard-of-hearing learners - primary users	Access complementary explanations, revisit relevant lesson content and practice supported ICT signs at their own pace
ICT teachers and special education instructors - supporting users	Review question attempts, repeated difficulties and learning histories to guide individual support
Schools and educational institutions - potential customers	Provide curriculum-focused learning assistance through a shared platform with configurable activities and notifications
Parents and guardians - indirect beneficiaries	Support home revision using the learner's selected topics and available learning information
Deaf-community organizations, sign-language specialists and educational partners	Contribute to vocabulary review, content development and evaluation of local suitability
Wider society	Potentially benefit from more inclusive access to ICT learning and opportunities for independent study

The suggested customer value is based upon consolidation of related support functions around a common curriculum and learner set. But institutional adoption

depends on content accuracy, compatibility with local sign-language, equipment availability, teacher preparation and continuing operating costs.

The benefits are based on documented functions and stakeholder needs. Stakeholder findings do not as yet exist in the evidence, therefore user acceptance, willingness to pay, reduced teaching workload and improved learning outcomes have not been established. Those findings would be necessary to establish whether the proposed value would lead to adoption.

7.2 Market alternatives and differentiation

The initial target segment for SignLearn AI is schools and special education programmers that support deaf and hard-of-hearing students studying Grade 10 and 11 ICT in Sri Lanka. Learners and teachers are the target users, whereas educational institutions and organizations that support them are the target buyers or implementers.

Available alternatives include learning management systems, AI-driven tutoring tools, sign-language translation products and teacher-developed educational resources. Table 7.2 compares the stated objectives of the alternatives with the one that the proposed solution can fulfill.

Table 7.2: Market Alternatives and Proposed Differentiation

Alternative	Existing value	SignLearn AI's proposed differentiation
Learning management systems, such as Moodle [5]	Deliver course content and assessments, with documented accessibility support.	Connect lesson timestamps and curriculum concepts to attention-informed revision, sign demonstrations and practice within the specified ICT learning context.
AI tutoring services, such as Khanmigo [6]	Provide conversational tutoring and teacher assistance.	Combine curriculum-focused explanations with available attention records, supported sign previews and tactile learning notifications.
Sign-language translation products, such as Signapse [7]	Provide AI-based translation into supported sign languages, including ASL and BSL.	Link a limited set of ICT sign demonstrations to lesson revision and guided practice. Local linguistic suitability still requires

		validation.
Teacher-prepared notes, captioned videos and sign demonstrations	Provide familiar, teacher-directed learning resources.	Associate these learning activities with shared topic context, recorded attempts and follow-up support.

Moodle documents course delivery, assessment and accessibility capabilities; Khan Academy describes Khanmigo as a tutor and teaching assistant; and Signapse identifies ASL and BSL among its supported translation languages. These comparisons relate to their documented purposes but do not confirm the unavailability of similar integrations.

SignLearn AI's most compelling proposed distinction centers on the coordination of its four components around Sri Lankan O/L ICT learning. Standalone features such as chatbots, quizzes and sign demonstrations are not unique offerings. A persuasive advantage would require validated curriculum mappings, locally reviewed sign resources, reliable integration and ongoing collaboration with educators and learners.

The project accordingly constitutes a specialized product proposition rather than one of proven market superiority. Its adoption and commercial viability require demonstration of user preference, institutional demand, operating costs and the comparative value of integration to existing resources.

7.3 Business model and feasibility

SignLearn AI suggests an institution-centric approach to adoption, with schools and special education programmers providing access to students and teachers. An initial limited adoption would then be used to scope the fit with curriculum, accessibility, technology requirements and support needs prior to wider adoption. Table 7.3 summarizes the proposed business and sustainability model.

Table 7.3: Proposed Business Model and Feasibility Considerations

Element	Proposed approach
Customers and users	Schools and educational organizations as potential customers; deaf and hard-of-hearing learners and ICT teachers as users
Channels	School demonstrations, teacher workshops and pilot programmers coordinated with educational and deaf-community organizations
Partners	ICT teachers, special education instructors, sign-language reviewers, educational organizations and hardware suppliers

Key resources	Maintained application code, curriculum content, validated gesture resources, hosting infrastructure, technical support and wristband devices
Main costs	Hosting, storage, language-model usage, content and sign review, software maintenance, teacher training and hardware assembly or replacement
Revenue model	Institutional subscriptions or support agreements, with optional wristband purchase or loan arrangements
Access and sustainability	Grants or sponsorship could subsidies access. Institutional support fees could contribute to maintenance and content updates.

The technical feasibility of this project is enabled by the web-based architecture, the defined interfaces and the documented capabilities and operations of components. The local fallback chatbot reduces the reliance on the deployed language model, though the availability of the backend and its data remains a requirement. The concurrent operation and physical delivery of notifications involves performance and hardware-related considerations and tests.

The operational feasibility involves computers, cameras, network equipment, teacher training and access to technical support. It also includes the ongoing operation and local review of sign language within the curriculum. The use of wristbands adds the considerations of charging, deployment and maintenance.

The financial feasibility of this concept has yet to be established by identifying costs and willingness to pay. A pilot would be needed to evaluate the cost of operations, including the number of supported students and the cost of hardware. Subscription-based payment options and expansion strategy would be informed by these results, with the current proposal only targeting an ongoing operation without relying on customers, funding or profitability.

7.4 Adoption intellectual property and compliance

The adoption of SignLearn AI relies on the availability of appropriate equipment, preparation of teachers, reliability of operation and trust in its ability to support learning. The schools would have to evaluate the availability of cameras, connection stability, maintenance requirements and relevance of supported signs before implementing it across classrooms. A staged approach with a pilot phase would enable addressing these concerns ahead of full-scale adoption.

Table 7.4: Adoption and Responsible Deployment Considerations

Area	Project consideration
Adoption barriers	Equipment costs, connectivity, teacher training and different communication preferences may affect uptake. Optional wristband use and clear onboarding can support flexible adoption.
Ownership	Ownership of original code, content, gesture assets and firmware must be clarified through contributor agreements and applicable university policies before commercialization.
Licensing	Third-party software, datasets, lesson materials and model assets require a licence inventory covering attribution, redistribution and commercial-use conditions.
Data protection	Student records and webcam-derived information require defined purposes, restricted access, retention limits and secure handling. Any information sent to external language-model services requires particular review.
Research ethics	Participation should be voluntary and supported by accessible information. Evaluation involving minors requires the institution’s applicable approval, guardian-consent and learner-assent procedures.
Safety and professional responsibility	Wristband comfort, electrical safety and vibration settings require testing. Automated observations and scores must be explained accurately and remain subject to appropriate human review.

WLASL's commercialization constraint is defined by their published terms which limit the data's use to academic and computational purposes and forbid commercial utilization [8]. It would therefore require appropriate permissions or alternative resources for a commercial release, with separate review of associated models and derived assets. Research access should not be assumed to qualify as commercial clearance.

The relevant Sri Lankan legal framework includes the Personal Data Protection Act, No. 9 of 2022, and its 2025 amendment [9]. Deployment planning must consider applicable provisions, commencement notices and requirements for external processing. This report cannot establish legal compliance through the mere presence of authentication or privacy-related design requirements.

Professional responsibility further requires accurate reporting of system limitations. Attention observations should not be assumed as definitive judgements of student ability, and supported sign demonstrations should not be presented as unrestricted

sign-language interpretation. Accessibility and linguistic suitability require review with the intended learner community.

7.5 Sustainable Development Goal contribution

SignLearn AI's main SDG target is 4: Quality Education with a focus on Target 4.5 promoting education for all - people with disabilities - and Target 4.a for inclusive learning environments [4]. The possible contribution of SignLearn AI is to overcome the barriers to ICT lessons and revision by providing additional texts, pictures and tactile information.

Table 7.5: Potential SDG Contributions, Risks and Sustainability Measures

SDG target	Plausible contribution	Possible negative effects	Measures to sustain benefits
4.5: Equal access to education	Transcripts, adaptive explanations and supported sign demonstrations provide alternative ways to engage with Grade 10 and 11 ICT content.	Equipment costs, connectivity requirements and limited sign coverage could exclude some learners.	Support existing school equipment, keep wristband use optional and expand locally reviewed content through educational partnerships.
4.a: Inclusive learning environments	Visual feedback, tactile notifications and learning records may help students participate and teachers provide relevant support.	Inaccurate attention interpretations, intrusive monitoring or confusing alerts could reduce trust and participation.	Provide clear camera controls, protect learner records, customize notifications and retain teacher oversight of automated indicators.

The features documented here provide a foundation for contributing to national SDG indicators, but do not directly indicate shifts in these metrics. The evaluation must centre on whether the learners can perform core activities autonomously, with appropriate explanations and without unneccessary barriers. Longitudinal analysis would be required to confirm improvements to learning outcomes or educational access.

The sustainability of improvements will depend on curriculum updates, sign-language reviews and educator training plus accessible technical support. Repair and responsible recycling of hardware is another concern, along with the environmental

and economic costs of video processing and language-model infrastructure. These factors should be taken into account to ensure the solution remains educationally viable without imposing unacceptable costs or risks.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

SignLearn AI responds to the shared problem of accessing both ICT lesson content and receiving relevant learning support when explanations are missed or challenging to understand. The project combines attention monitoring and knowledge graph questions, an adaptive chatbot system, sign-avatar-based demonstrations, and wristband-assisted sign practice, within a single Grade 10 and 11 ICT learning context.

The primary contribution of the project is the interconnectivity of these functions. The provision of lesson timestamps and curriculum mappings inform concept-related questioning and revision suggestions. Learners are able to ask for further explanations, view relevant sign demonstrations and practice related vocabulary, while record of learning supports teacher guidance. In this way, the project offers a comprehensive approach to both accessing relevant content and receiving relevant learning support.

The provided implementation example supports the conclusion that the four proposed research contributions have been organised around a shared learning workflow. However, the project does not yet establish the reliability of all integrations, or demonstrate improved learning outcomes, stakeholder acceptance or suitability for the target population. Attention observations are indirect, sign support is limited to a small vocabulary, and physical notifications require device connectivity.

The shared problem of accessing both ICT lesson content and receiving relevant learning support is thus addressed at the level of system design and documented implementation, with educational effectiveness remaining to be established. SignLearn AI offers a valuable foundation for accessible, curriculum-relevant learning support, the effectiveness of which depends on accurate content, locally-appropriate signs, dependable functionality and evaluation with stakeholders.

8.2 Principal integrated contribution

The main contribution of SignLearn AI is a curriculum-linked learning workflow that combines participation in lessons, revision, demonstration of signs and guided practice. This has wider value than its individual components by enabling information generated during one activity to support a different one. A timestamp from a lesson can identify a concept to be revised, which can in turn guide a chatbot explanation, and vocabulary support can link this to a relevant sign demonstration and practice activity.

The team have created a shared application structure, integration interfaces and curriculum mappings that enable this coordination. These combine to form a reusable design for textual, visual and tactile assistance within a subject-specific learning environment. The common report also establishes evaluation criteria that

differentiate between successful individual component operation and end-to-end support.

A key integration lesson is that shared meaning is as important as technical connections. Components need to have a common understanding of the student, lesson, concept and sign being referred to, and the limitations of the information being shared. Missing attention data, uncertain practice feedback and unconfirmed notification delivery must be distinguished from valid learning outcomes.

The contribution is therefore both an integrated implementation and an evaluation framework. It forms a foundation for further research into accessible learning support, but its application to other subjects or learner groups would require new content mappings, linguistic review and user evaluation.

8.3 Future directions

Future work should first consolidate the basis of the integrated platform before extending its functionality. Immediate priorities are to confirm full learning workflows, and to assess them with deaf and hard-of-hearing learners, ICT teachers and sign-language reviewers. Table 8.1 shows the proposed priorities.

Table 8.1: Priorities for Future Development and Evaluation

Priority	Future work	Intended outcome
1. Verify integrated operation	Test curriculum mappings, record consistency, service failures, access controls and physical wristband delivery under documented conditions.	Reproducible evidence of reliable end-to-end behavior
2. Validate learner suitability	Conduct accessible task-based evaluations and review supported signs with the local learner community.	Evidence of usability, linguistic suitability and necessary improvements
3. Assess educational value	Compare equivalent learning activities with and without integrated support, using direct knowledge assessments.	Evidence of whether the platform improves understanding or retention
4. Prepare deployment	Establish monitoring, backup restoration, configuration management and maintenance responsibilities. Resolve licensing restrictions affecting commercial use.	A maintainable deployment suitable for a controlled institutional pilot

5. Evaluate adoption and scaling	Measure concurrent-session performance, operating costs, teacher support needs and institutional interest.	Realistic capacity limits and a sustainable adoption model

Further research could investigate how explicit learner feedback and question performance can be used to improve the interpretation of attention observations. Sign practice should also be investigated across different users, camera conditions and signing variations before generalizing its feedback.

Expansion to further vocabulary, subjects or language support should be considered following successful validation of the current ICT scope. This sequence emphasizes reliable operation and proven learner value while providing a pragmatic path to wider adoption.

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